

# OTTAVIA PERSONENI

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## EDUCATION

### Stanford University

Candidate for MS in Aerospace Engineering

June 2027

Stanford, CA

### Massachusetts Institute of Technology (MIT)

BS Degree in Mechanical Engineering with Computer Science

May 2025

Cambridge, MA

- **GPA:** 4.9/5.0

## EXPERIENCE

### MIT Portugal Marine Robotics Summer School

July 2025

1 of 12 MIT undergrad/grad/PhD students selected

Horta, Faial, Azores, Portugal

- MIT Portugal program with lectures and fieldwork on marine robotics, oceanography, and ecosystem monitoring.
- Collaborated with teammates, professors, field experts, and the local community to design and operate an ROV for tracking invasive algae around Faial Island.

### MIT Space Propulsion Lab (SPL) - Electrospray Thruster Characterization

September 2024 – May 2025

Undergraduate Researcher

Cambridge, MA

- Documenting, characterizing, and developing a consistent fabrication process for electrospray emitters.
- TVAC testing emitters using novel materials and unique propellants to achieve higher performance reliability.

### MIT Mechanical Engineering Capstone Course

September 2024 – December 2024

Sub Team Lead and Presenter

Cambridge, MA

- 2.009 (Product Engineering Process): led the electronics and software sub team in building a novel underwater communication device for scuba divers.
- Chosen as the final presenter to display our product, NepΨune, to an online and in person audience of 15,000 people.

### Firefly Aerospace – Blue Ghost (Lunar Delivery & Operations)

May 2024 - August 2024

Spacecraft and Avionics Intern

Austin, Texas

- Built and programmed a high-fidelity simulation testbed to verify lunar lander commands. Created modules to simulate 90+ heater power draws and high/ambient/low temperature settings for 120+ temperature sensors.
- Automated the electronics selection process for the heater circuits, used a network of Raspberry Pis to control inputs and relays, soldered over a thousand electrical connections, and created a GUI to test temperature sensor set points.
- Designed and installed a Firefly plaque onto the lunar lander (which has successfully landed on the Moon!)
- First intern to receive a monthly Blue Ghost program wide award for interdisciplinary contributions.

### IMDEA Energy – High Temperature Processes Unit

June 2023 - August 2023

Computational Fluid Dynamics (CFD) Engineering Intern

Madrid, Spain

- Modeled fluid flow, heat transfer, and melt fraction trends of phase-change capsules in latent thermal energy storage systems (Green Hydrogen project).
- Analyzed capsule characteristics and efficiency, contributing findings to a departmental research paper.

### D-Orbit

June 2022 - August 2022

Aerospace Engineering Intern

Fino Mornasco (CO), Italy

- Programmed an Arduino to remotely control satellite components, and tested capabilities on satellite EM.
- Created a GUI that included a satellite status dashboard, a computer data to satellite records transfer, and a proprietary dispenser testing system.

## LEADERSHIP AND SERVICE

### MIT First Year Student Award

May 2022

- 1 out of 5 first-year students awarded in 2022 for remarkable achievements.

### MIT Women's Varsity Volleyball

September 2021 – May 2025

Captain & Libero

Cambridge, MA

- Led team to three consecutive NEWMAC Championships and two NCAA DIII Elite 8 finishes.
- Honors: Academic All-American First Team ('22), NEWMAC Defensive Player of the Year ('22, '24) & Rookie of the Year ('21), DIII All-American Honorable Mention ('21- '24), All-Conference & All-Region First Team ('21- '24), 13-time NEWMAC Player of the Week ('21- '24).

### MIT Sloan Business Club (SBC) Social Chair

December 2021 – January 2023

MIT Alpha Phi House Manager

October 2021 – May 2024

## SKILLS AND INTERESTS

- **Languages:** English (native) | Italian (native) | Spanish (fluent)
- **Skills:** COMSOL | Python | C++ | Github | Matlab | Excel
- **Volunteering:** DC Central Kitchen, Fondazione Rava & Nuestros Pequeños Hermanos (Honduras)